Methods in Computational Linguistics I — HW 5
Example Program Runs and Outputs

These examples show different cases of running the completed program.

1 Searching games by console and release year

```
$ python3 HW5-search.py
This program helps the user explore a data set of
over a thousand video games released between
2004 and 2008.

In particular, you can:
1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 1
Enter year: 2005
Enter console: X360

                          TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------
                       Amped 3     72  0.13 14.95    X360 2005      9.0
  Battlefield 2: Modern Combat     77  0.36 16.95    X360 2005      6.9
                Call of Duty 2     89  1.69 17.95    X360 2005      9.0
   Condemned: Criminal Origins     81  0.33 14.95    X360 2005      7.5
              Dead or Alive 4     85  0.38 17.95    X360 2005      6.0
                          GUN     75  0.22 19.95    X360 2005      8.0
      Kameo: Elements of Power     79  0.31 11.95    X360 2005     10.2
                Madden NFL 06     74  0.62  4.95    X360 2005      3.8
                       NBA 2K6     81  0.22  4.95    X360 2005      0.5
                  NBA LIVE 06     64  0.29  4.95    X360 2005      4.0
   Need for Speed: Most Wanted     83  0.95 17.95    X360 2005     12.0
                       NHL 2K6     75  0.10  5.95    X360 2005      2.5
             Perfect Dark Zero     79  0.70 17.95    X360 2005      8.0
Peter Jackson's King Kong: The     80  0.32 14.95    X360 2005      7.0
                       Quake 4     75  0.27 14.95    X360 2005      8.4
                Ridge Racer 6     74  0.10 14.95    X360 2005      8.0
       Tiger Woods PGA Tour 06     71  0.33  9.95    X360 2005     -1.0
Tony Hawk's American Wasteland     75  0.29 14.95    X360 2005     13.0

18 matches found

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 1
```

```
Enter year: 2006
Enter console: Wii

                          TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------
                 Call of Duty 3     69  1.15 19.95     Wii 2006      8.4
Dragon Ball Z: Budokai Tenkaic      72  0.24 14.95     Wii 2006     14.6
                       Elebits      75  0.22 14.95     Wii 2006      9.0
                   Excite Truck     72  0.40 19.95     Wii 2006     14.0
             Far Cry: Vengeance     38  0.07 16.95     Wii 2006     -1.0
                    Happy Feet     46  0.13 14.95     Wii 2006      0.7
       Ice Age 2: The Meltdown     66  0.07 17.95     Wii 2006      8.5
       Kororinpa: Marble Mania     69  0.08 17.95     Wii 2006     -1.0
                  Madden NFL 07     81  0.48  9.95     Wii 2006     -1.0
      Marvel Ultimate Alliance     73  0.38 17.95     Wii 2006     14.0
          Metal Slug Anthology     73  0.11 16.95     Wii 2006      7.0
    Monster 4x4: World Circuit     51  0.82 12.95     Wii 2006     -1.0
        Need for Speed: Carbon     67  0.47 17.95     Wii 2006     13.6
                  Open Season     49  0.07 17.95     Wii 2006      3.8
      Pokmon Battle Revolution     53  0.81 39.95     Wii 2006     15.4
    Rampage: Total Destruction     46  0.19 16.95     Wii 2006      5.0
         Rayman Raving Rabbids     76  1.26 17.95     Wii 2006      6.2
                    Red Steel     63  0.58 14.95     Wii 2006      8.2
SpongeBob Squarepants: Creatur     57  0.44 14.95     Wii 2006      6.8
Super Monkey Ball: Banana Blit     74  1.00 14.95     Wii 2006      5.4
              Super Swing Golf     72  0.14 16.95     Wii 2006     -1.0
                 The Ant Bully     45  0.09 16.95     Wii 2006      3.6
The Legend of Zelda: Twilight      95  3.33 39.95     Wii 2006     43.0
Tom Clancy's Splinter Cell: Do     61  0.11 17.95     Wii 2006     11.0
 Trauma Center: Second Opinion     80  0.28 17.95     Wii 2006      9.0
       WarioWare: Smooth Moves     83  0.90 39.95     Wii 2006      3.0
                     Wii Play     58 14.66 14.95     Wii 2006      4.0

27 matches found

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 1
Enter year: 2004
Enter console: Wii

Sorry, no matches were found.

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 6
Goodbye!
```

2 Searching games by title

```
$ python3 HW5-search.py
This program helps the user explore a data set of
over a thousand video games released between
2004 and 2008.

In particular, you can:
1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 2
Enter game title (or start of title): Sonic
                          TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------
    Sonic and the Secret Rings     69  1.08 16.95     Wii 2007      9.3
Sonic Chronicles: The Dark Bro     74  0.52 17.95      DS 2008     20.0
    Sonic Riders: Zero Gravity     56  0.45 16.95     Wii 2008      4.3
                  Sonic Rivals     64  0.76 17.95     PSP 2006      6.5
                Sonic Rivals 2     60  0.46 22.95     PSP 2007     22.0
                    Sonic Rush     82  1.20 14.95      DS 2005      6.2
          Sonic Rush Adventure     78  0.52 24.95      DS 2007      9.0
            Sonic the Hedgehog     46  0.41 14.95    X360 2007      3.1
            Sonic the Hedgehog     43  0.40 16.95     PS3 2007      3.1
               Sonic Unleashed     66  1.09 17.95     Wii 2008     13.0
               Sonic Unleashed     54  0.42 17.95     PS3 2008     13.0
               Sonic Unleashed     60  0.41 17.95    X360 2008     13.0

12 matches found

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 2
Enter game title (or start of title): frogg
                          TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------
         Frogger: Helmet Chaos     66  0.11 19.95     PSP 2005     -1.0
         Frogger: Helmet Chaos     69  0.06 17.95      DS 2005     -1.0

2 matches found

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 2
```

```
Enter game title (or start of title): CS21
Sorry, no matches were found.

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 6
Goodbye!
```

3 Checking for valid input

```
$ python3 HW5-search.py
This program helps the user explore a data set of
over a thousand video games released between
2004 and 2008.

In particular, you can:
1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 0

Invalid choice, try again.

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 5

Invalid choice, try again.

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: -6

Invalid choice, try again.

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
```

```
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: title

Invalid choice, try again.

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 1
Enter year: 1999
Please enter a year between 2004-2008.
Enter year: 2005
Enter console: PS3

Sorry, no matches were found.

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 6
Goodbye!
```

4 Getting a sorted list of games by review score

```
$ python3 HW5-search.py
This program helps the user explore a data set of
over a thousand video games released between
2004 and 2008.

In particular, you can:
1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 5

List of game genres in the data set:
-----------------------------------
Action
Adventure
Driving
Educational
Racing
```

```
Role-Playing(RPG)
Simulation
Sports
Strategy

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 3
Enter console (or 'Any'): Wii
Enter genre (or 'Any'): Sports

                         TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------
           MLB Power Pros 2008     83  0.12 22.95     Wii 2008     -1.0
     Pro Evolution Soccer 2008     83   0.1 17.95     Wii 2008     70.0
       WarioWare: Smooth Moves     83   0.9 39.95     Wii 2006      3.0
                  Madden NFL 07     81  0.48  9.95     Wii 2006     -1.0
                        Wii Fit     80   9.6 24.95     Wii 2007      4.0
        Mario Strikers Charged     79  0.93 29.95     Wii 2007     12.0
Shaun White Snowboarding: Road     78  0.95 14.95     Wii 2008     -1.0
                  Madden NFL 08     76  0.86 12.95     Wii 2007     17.0
                       SSX Blur     74  0.31 17.95     Wii 2007     10.0
Rayman Raving Rabbids TV Party     73  0.74 17.95     Wii 2008     -1.0
              Super Swing Golf     72  0.14 16.95     Wii 2006     -1.0
       Tiger Woods PGA Tour 08     72     1 17.95     Wii 2007     -1.0
        Sega Superstars Tennis     71  0.28 12.95     Wii 2008      8.0
       Tiger Woods PGA Tour 07     71  0.46 17.95     Wii 2007     10.0
                       skate it     70   0.5 39.95     Wii 2008     -1.0
     Super Swing Golf Season 2     70  0.07 19.95     Wii 2007     -1.0
Mario & Sonic at the Olympic G     67  2.74 29.95     Wii 2007      9.0
                  EA Playground     66  0.72 17.95     Wii 2007     -1.0
                       The BIGS     66  0.36 17.95     Wii 2007    10.13
                        NHL 2K9     65  0.06 14.95     Wii 2008     -1.0
                      Surf's Up     64  0.16 16.95     Wii 2007     1.28
             SEGA Bass Fishing      59  0.34 17.95     Wii 2008      7.0
 Victorious Boxers: Revolution     59  0.08 24.95     Wii 2007     -1.0
    WWE Smackdown vs. Raw 2008     59  0.39 12.95     Wii 2007     13.1
                       Wii Play     58 14.66 14.95     Wii 2006      4.0

25 matches found

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 3
Enter console (or 'Any'): Any
Enter genre (or 'Any'): Any

                         TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
```

```
------------------------------ ------ ----- ----- ------- ---- --------
           Grand Theft Auto IV     98  5.76 24.95    X360 2008     32.0
           Grand Theft Auto IV     98  3.91 24.95     PS3 2008     32.0
            Super Mario Galaxy     97  4.94 39.95     Wii 2007     20.0
                     BioShock      96  1.52 16.95    X360 2007     13.5
                The Orange Box     96  1.06 17.95    X360 2007     29.0
               LittleBigPlanet     95  2.46 24.95     PS3 2008     10.0
The Legend of Zelda: Twilight      95  3.33 39.95     Wii 2006     43.0
                     BioShock      94  0.53 17.95     PS3 2008     13.5
Call of Duty 4: Modern Warfare     94  5.77 29.95    X360 2007      7.0
Call of Duty 4: Modern Warfare     94  2.91 24.95     PS3 2007      7.0
                  Gears of War     94   3.7 19.95    X360 2006     9.43
                       Halo 3      94  8.03 24.95    X360 2007     9.08
Metal Gear Solid 4: Guns of th     94  2.58 22.95     PS3 2008     18.9
The Elder Scrolls IV: Oblivion     94  2.12 19.95    X360 2006     80.0
                    Fallout 3      93  2.44 17.95    X360 2008     45.0
                Gears of War 2     93   3.9 24.95    X360 2008     10.0
      Super Smash Bros.: Brawl     93  5.71 39.95     Wii 2008     15.0
The Elder Scrolls IV: Oblivion     93   1.2 17.95     PS3 2007     80.0
                Chrono Trigger     92  0.51 17.95      DS 2008     25.0
                Guitar Hero II     92  2.04 17.95    X360 2007      9.5
                     Rock Band     92  2.02 14.95    X360 2007      8.0
                     Rock Band     92  1.01 14.95     PS3 2007      8.0
                     Rock Band     92   1.4 17.95     Wii 2008      8.0
                   Rock Band 2     92  1.85 19.95    X360 2008     12.5
 God of War: Chains of Olympus     91  1.54 17.95     PSP 2008     5.67

25 matches found

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 3
Enter console (or 'Any'): Any
Enter genre (or 'Any'): Racing

                         TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------
           Grand Theft Auto IV     98  5.76 24.95    X360 2008     32.0
           Grand Theft Auto IV     98  3.91 24.95     PS3 2008     32.0
                 Mario Kart DS     91  9.99 29.95      DS 2005     9.08
            Forza Motorsport 2     90  2.48 12.95    X360 2007    58.22
Grand Theft Auto: Liberty City     88  2.97 17.95     PSP 2005    14.97
                   Ridge Racer     88  0.36 19.95     PSP 2004     0.88
                 WipEout Pure      88  0.45 12.95     PSP 2005      8.8
              Burnout Paradise     87  1.01 17.95     PS3 2008     20.0
                          GRID     87  0.35 19.95    X360 2008    23.13
                          GRID     87  0.31 24.95     PS3 2008    23.13
               Burnout Legends     86  0.68 14.95     PSP 2005     12.0
Grand Theft Auto: Vice City St     86  1.72 17.95     PSP 2006    17.23
                          Pure     85  1.43 12.95    X360 2008     13.0
                     Crackdown     83  1.02 14.95    X360 2007    12.33
                          DiRT     83  0.39 17.95    X360 2007     13.0
                          DiRT     83  0.16 17.95     PS3 2007     13.0
```

```
   Need for Speed: Most Wanted     83  0.95 17.95    X360 2005     12.0
               No More Heroes      83  0.29 17.95     Wii 2007     13.0
                         Pure      83  0.36 17.95     PS3 2008     13.0
       WarioWare: Smooth Moves     83   0.9 39.95     Wii 2006      3.0
               Mario Kart Wii      82 12.39 29.95     Wii 2008     9.12
    Midnight Club: Los Angeles     82  1.09 22.95     PS3 2008    31.03
                   MotorStorm     82  1.58 16.95     PS3 2006     15.0
      MotorStorm: Pacific Rift     82  0.37 29.95     PS3 2008     32.5
    OutRun 2006: Coast 2 Coast     82  0.03 17.95     PSP 2006      4.0

25 matches found


1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 3
Enter console (or 'Any'): Wii
Enter genre (or 'Any'): Role-Playing (RPG)

                         TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------
The Legend of Zelda: Twilight      95  3.33 39.95     Wii 2006     43.0
             Super Paper Mario     85   1.5 29.95     Wii 2007    19.77
     Fire Emblem: Radiant Dawn     78  0.26 34.95     Wii 2007    47.83
Final Fantasy Fables: Chocobo'     76  0.09 17.95     Wii 2008     25.0
      Marvel Ultimate Alliance     73  0.38 17.95     Wii 2006     14.0
               Dokapon Kingdom     72  0.12 17.95     Wii 2008     25.0
  Harvest Moon: Magical Melody     69  0.18 23.95     Wii 2008    30.37
Tales of Symphonia: Dawn of th     68   0.2 24.95     Wii 2008    36.77
Harvest Moon: Tree of Tranquil     65  0.62 34.95     Wii 2008     38.5
The Chronicles of Narnia: Prin     63  0.33 16.95     Wii 2008    11.65
      Neopets Puzzle Adventure     52  0.14 14.95     Wii 2008      3.0
                      Baroque     50  0.04 19.95     Wii 2008     7.78

12 matches found

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 3
Enter console (or 'Any'): X360
Enter genre (or 'Any'): Computer Science

                         TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------

0 matches found

1. Search by console and release year
```

```
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 6
Goodbye!
```

5 Getting a sorted list of games by sales figures

```
$ python3 HW5-search.py
This program helps the user explore a data set of
over a thousand video games released between
2004 and 2008.

In particular, you can:
1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 4
Enter year: 2005

                          TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------
                 Mario Kart DS     91  9.99 29.95      DS 2005     9.08
                    Nintendogs     83  9.72 26.95      DS 2005    27.98
Brain Age: Train Your Brain in     77  5.07 17.95      DS 2005      1.0
Brain Agey: More Training in M     77  3.65 17.95      DS 2005     15.0
Grand Theft Auto: Liberty City     88  2.97 17.95     PSP 2005    14.97
   Animal Crossing: Wild World     86   2.6 24.95      DS 2005    80.53
             Big Brain Academy     74  1.77 16.95      DS 2005     7.13
                Call of Duty 2     89  1.69 17.95    X360 2005      9.0
  Midnight Club 3: DUB Edition     74  1.69 14.95     PSP 2005     48.0
Need for Speed: Most Wanted 5-     72  1.67 14.95     PSP 2005    10.33
Pokmon Mystery Dungeon: Blue R     62  1.22 14.95      DS 2005    25.62
                    Sonic Rush     82   1.2 14.95      DS 2005     6.22
     Star Wars: Battlefront II     69  1.09 16.95     PSP 2005      9.0
SOCOM: U.S. Navy SEALs - Firet     82  1.07 14.95     PSP 2005     8.58
   Need for Speed: Most Wanted     83  0.95 17.95    X360 2005     12.0
                 Zoo Tycoon DS     44  0.92 12.95      DS 2005     -1.0
                    The Sims 2     70  0.83 17.95      DS 2005    34.92
                 Madden NFL 06     75  0.75  7.95     PSP 2005     3.75
Mario & Luigi: Partners in Tim     86  0.75 29.95      DS 2005    18.65
Need for Speed Underground: Ri     74  0.73 17.95     PSP 2005    20.47
        Twisted Metal: Head-On     79  0.73 14.95     PSP 2005     -1.0
             Perfect Dark Zero     79   0.7 17.95    X360 2005      8.0
Super Monkey Ball: Touch & Rol     63   0.7 15.95      DS 2005     0.57
          Super Princess Peach     75   0.7 29.95      DS 2005     9.28
               Burnout Legends     86  0.68 14.95     PSP 2005     12.0

25 matches found
```

```
1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 4
Enter year: 2008

                         TITLE REVIEW SALES PRICE CONSOLE YEAR PLAYTIME
------------------------------ ------ ----- ----- ------- ---- --------
                Mario Kart Wii     82 12.39 29.95     Wii 2008     9.12
           Grand Theft Auto IV     98  5.76 24.95    X360 2008     32.0
      Super Smash Bros.: Brawl     93  5.71 39.95     Wii 2008     15.0
    Call of Duty: World at War     84  4.58 29.95    X360 2008      8.0
           Grand Theft Auto IV     98  3.91 24.95     PS3 2008     32.0
                Gears of War 2     93   3.9 24.95    X360 2008     10.0
      Pokmon: Platinum Version     83  2.86 27.95      DS 2008     43.0
Metal Gear Solid 4: Guns of th     94  2.58 22.95     PS3 2008     18.9
                      Fable II     89  2.55 17.95    X360 2008     20.0
    Call of Duty: World at War     85  2.53 29.95     PS3 2008      8.0
       Guitar Hero: World Tour     86  2.47 17.95     Wii 2008     8.62
              LittleBigPlanet     95  2.46 24.95     PS3 2008     10.0
                    Fallout 3     93  2.44 17.95    X360 2008     45.0
LEGO Indiana Jones: The Origin     77  2.42  9.95    X360 2008    13.72
                  Left 4 Dead     89  2.39 19.95    X360 2008      7.0
                Madden NFL 09     83  2.33 14.95    X360 2008    14.52
          Guitar Hero: On Tour     71  2.22 14.95      DS 2008     24.0
                Kung Fu Panda     75  2.04  9.95    X360 2008      5.0
                  Rock Band 2     92  1.85 19.95    X360 2008     12.5
        Sega Superstars Tennis     67  1.84  9.95    X360 2008      8.0
    Animal Crossing: City Folk     73  1.78 29.95     Wii 2008     65.0
       Guitar Hero: World Tour     85  1.78 16.95    X360 2008     8.62
Mario & Sonic at the Olympic G     70  1.74 24.95      DS 2008      9.0
                Madden NFL 09     85  1.66 14.95     PS3 2008    14.52
    LEGO Batman: The Videogame     76  1.64 14.95    X360 2008     15.0

25 matches found

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 2009
Invalid choice, try again.

1. Search by console and release year
2. Search by game title
3. Rank best reviewed games by console and genre
4. Find top selling games by year
5. Help: list possible Genres
6. Quit

Choice: 6
```

```
Goodbye!
```